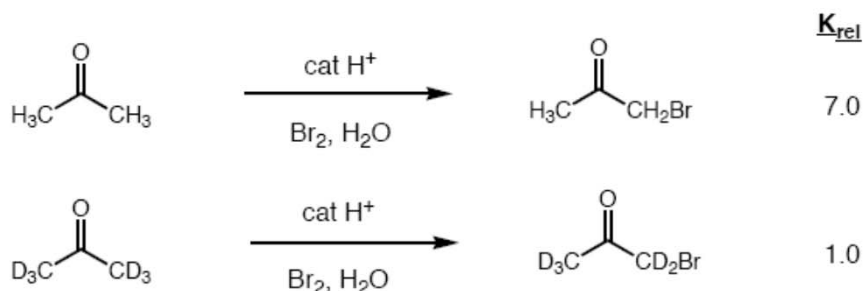


What Are Kinetic Isotope Effects?

- A kinetic isotope effect is a mechanistic phenomenon wherein isotopically substituted molecules react at different rates.



Key Assumptions

- Isotopic substitution does not affect the potential energy surface of the reaction or the energies of the electronic states.
- Only mass dependent properties are affected, most importantly vibrational frequencies.

- Interpretation of the rate differences provides information on the nature of the rate-determining step.



- There are several different classifications for KIE's

Primary isotope effect: Occurs when labelled bond is made or broken in the rate determining step

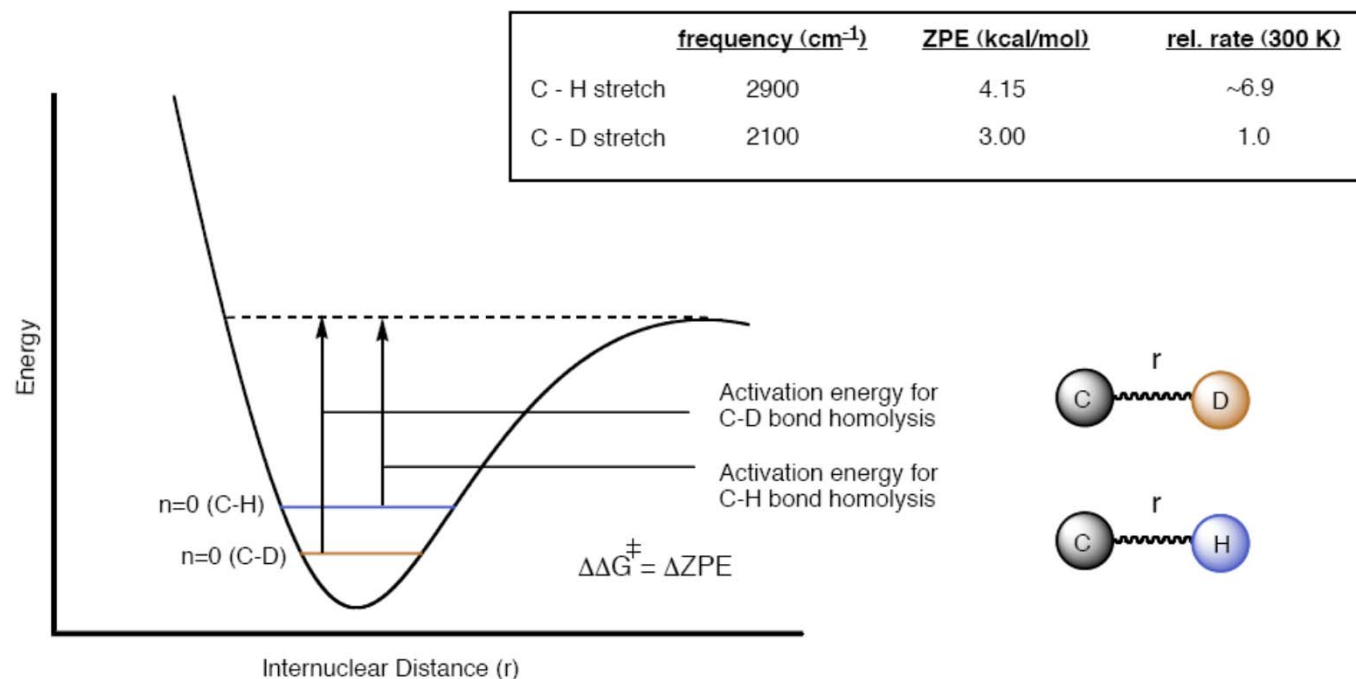
Secondary isotope effect: Occurs when labelled bond is not made or broken in the rate determining step

Normal isotope effect: Occurs when K_H/K_D is greater than 1

Inverse isotope effect: Occurs when K_H/K_D is less than 1

Physical Origins of Primary Kinetic Isotope Effects

- Simplest case to treat is the homolytic cleavage of C - H/D bond where the bond is considered to be fully broken at the transition state. Reaction progress followed by observing the C - H/D bond stretch.

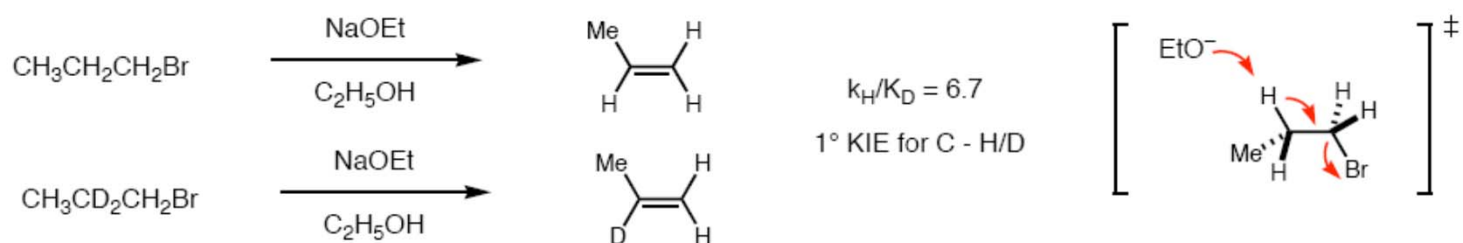


- As the C - H/D bond breaks at the transition state the stretch becomes a translation. As a result there is no new stretch in the TS that corresponds to the stretch of ground state bond. For this mechanism, the isotope effect is entirely controlled by the difference in the ground state ZPE's.

Using Primary KIE's to Distinguish Between Reaction Mechanisms

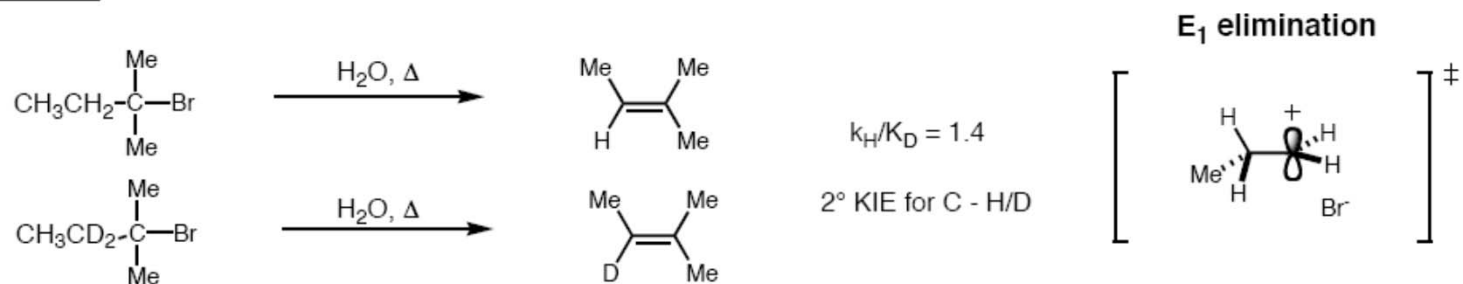
- KIE's give useful information about the rate determining step of the reaction mechanism

Case 1



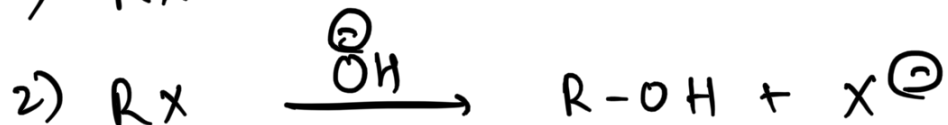
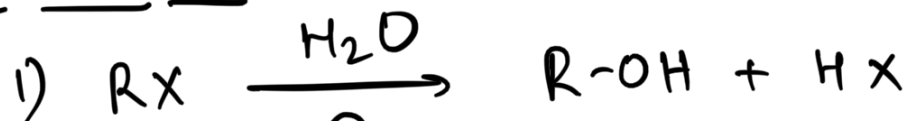
- This KIE is consistent with an E₂ elimination in which the C - H/D bond is broken in the rate determining step.

Case 2



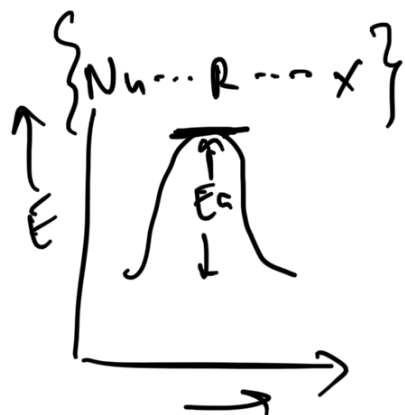
- This KIE is consistent with an E₁ solvolysis in which the C - H/D bond is not broken in the rate determining step.

② Kinetic method



1) Rate = $k[\text{RX}]$ unimolecular $\text{S}_{\text{N}}1$ pseudomolecular
R.D.S. does not involve H_2O

2) Rate = $k[\text{RX}][\text{OH}^-]$ $\text{S}_{\text{N}}2$ bimolecular
↓
Nucleophile

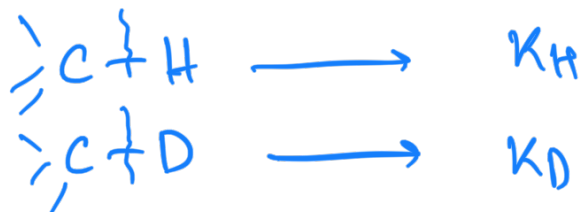


② Isotope labelling $\rightarrow$ kinetic isotope effect (K.I.E.)
 $\rightarrow$ isotopic atom replacement not kinetic

* To determine R.D.S. in the reaction

slowest step $\rightarrow$ KIE

* Replace (H) by heavier isotope i.e. (D)



$$\frac{k_H}{k_D} \approx 1.2$$

C-H bond
cleavage is
not the
R.D.S

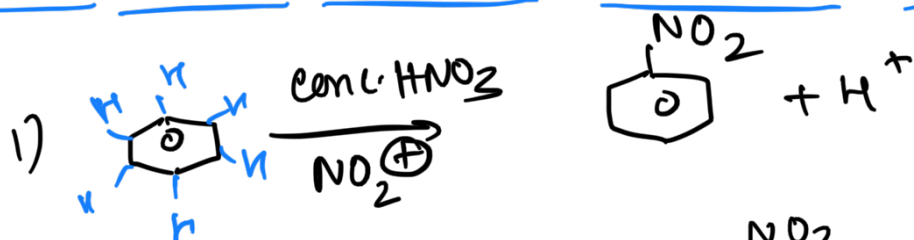
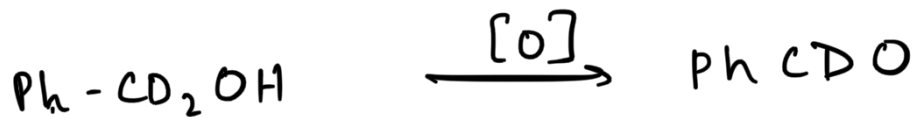
C-H cleavage
is involved in RDS

$$\frac{k_H}{k_D} > 1 \quad 3-20$$

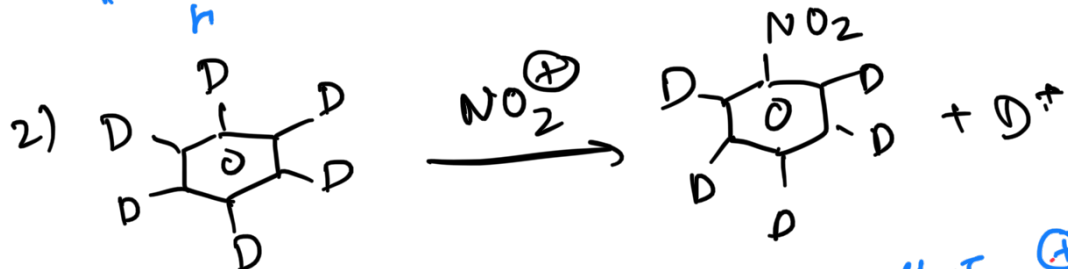
Primary K.I.E.



$$\frac{k_H}{k_D} = 7 \quad \text{Primary k.i.E.}$$

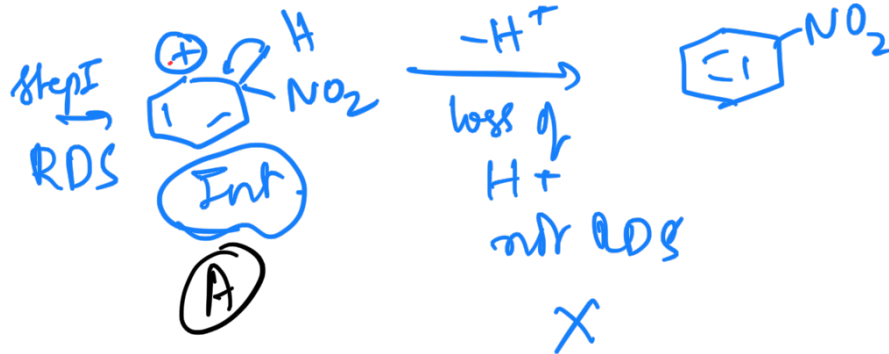


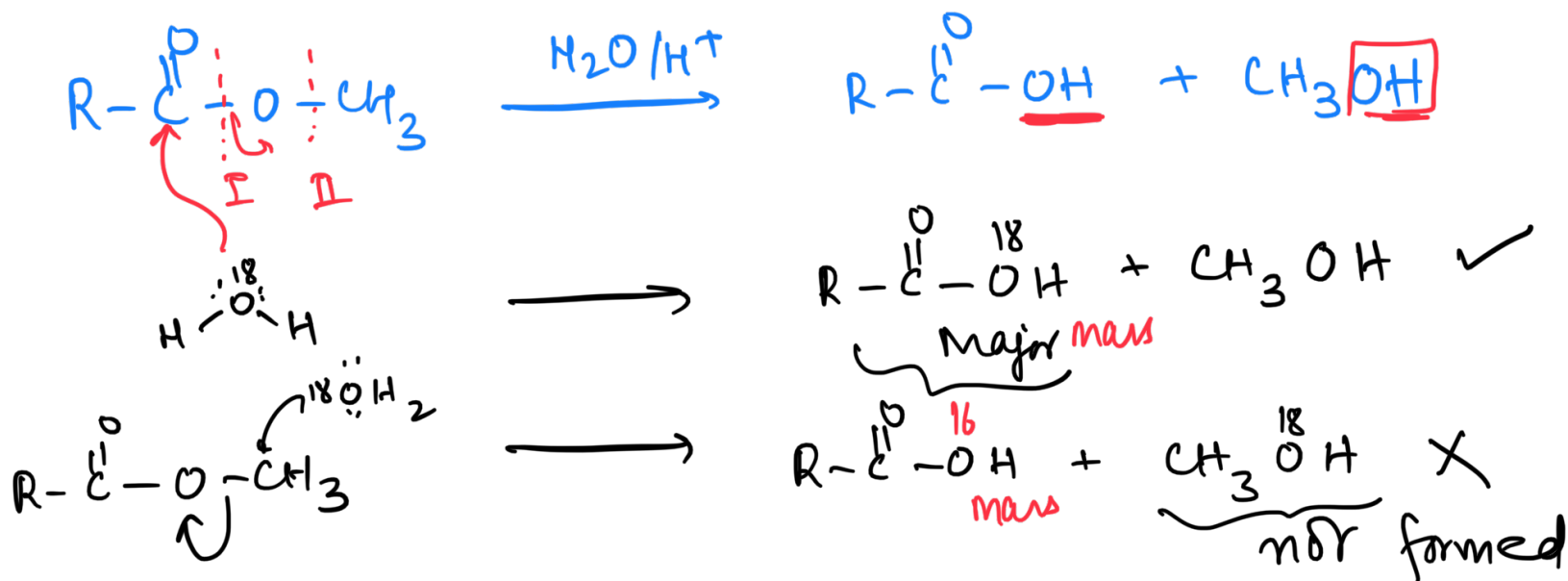
$$\frac{k_H}{k_D} \approx 1$$



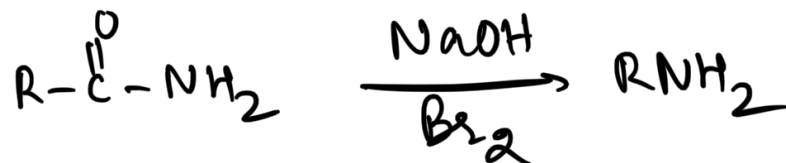
C-H/E-D

Electrophilic sub. of benzene

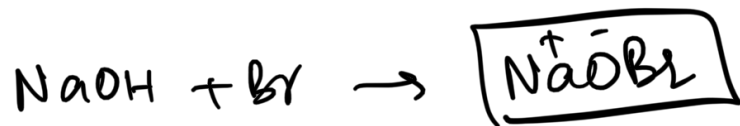


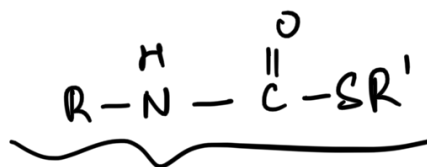
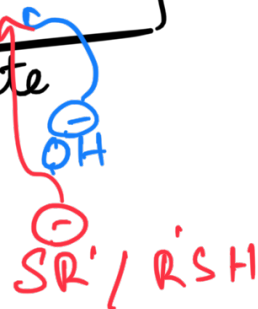
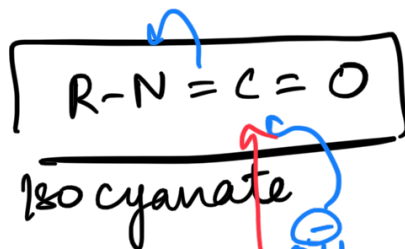
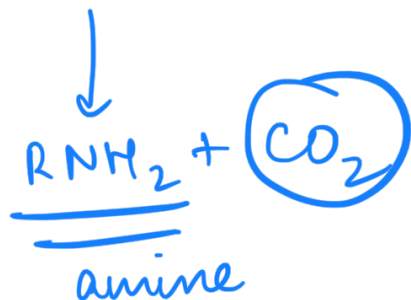
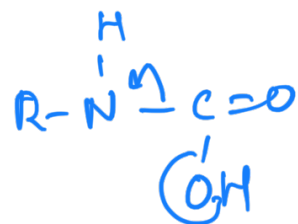
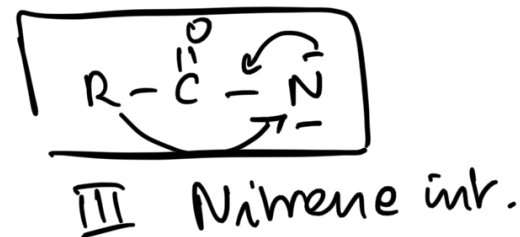
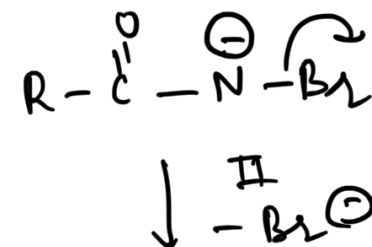
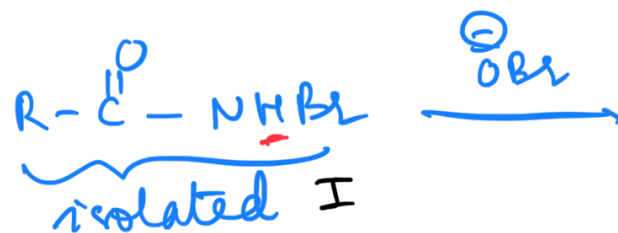
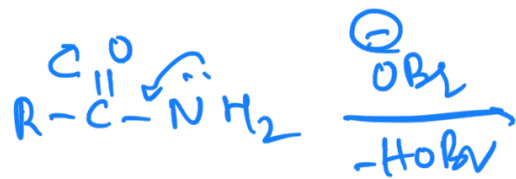


④ Identification of Intermediates



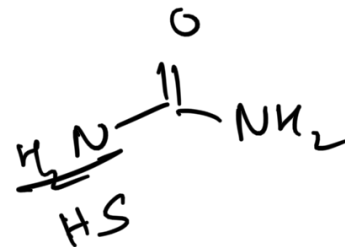
Hoffman bromamide
 degradation
 = 1C less -



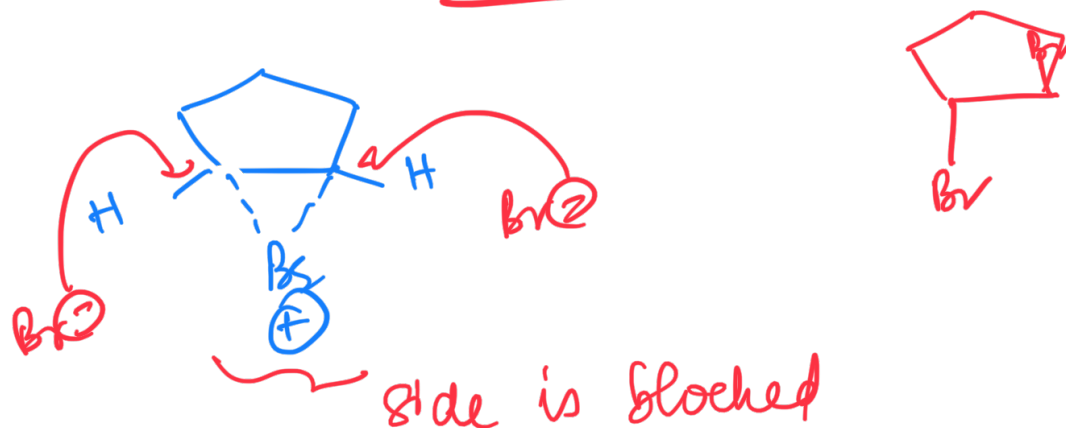
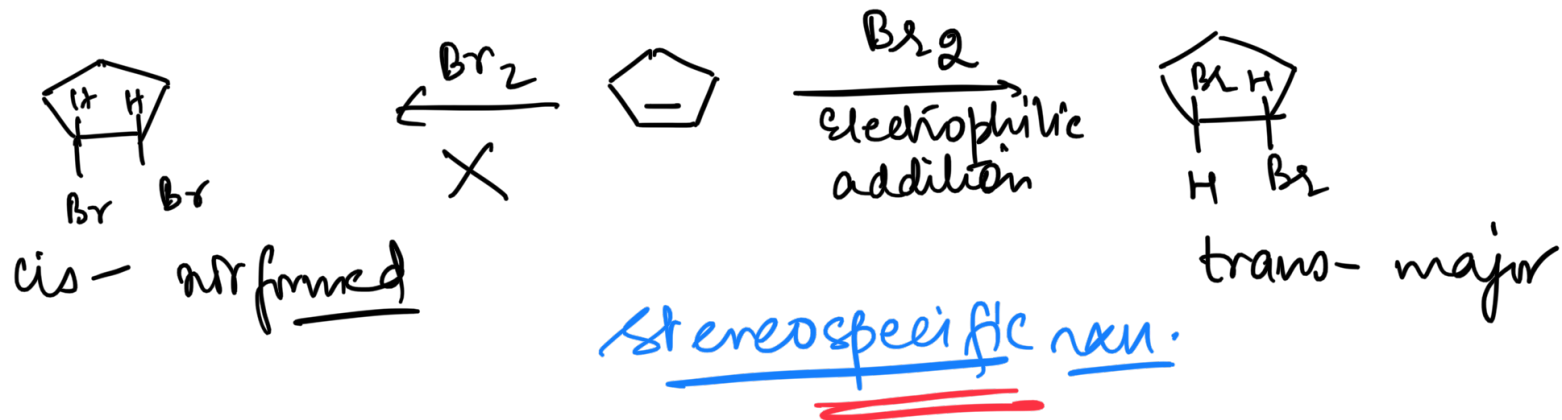


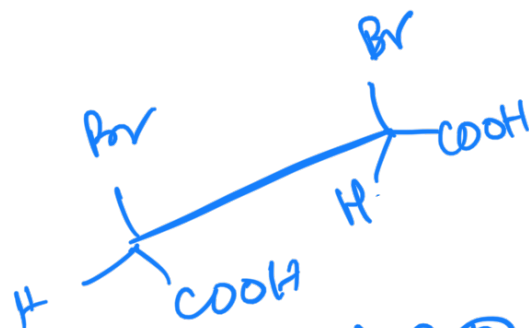
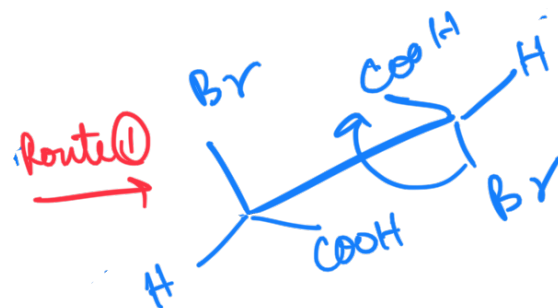
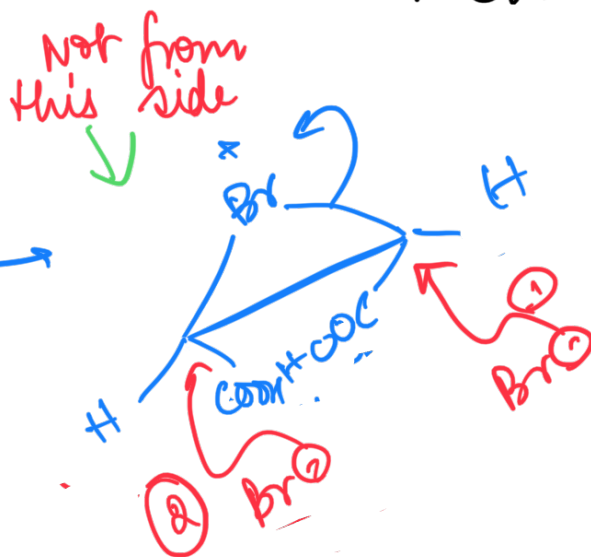
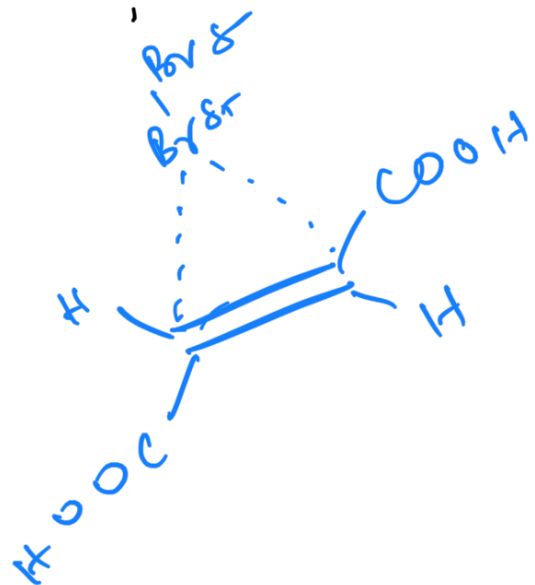
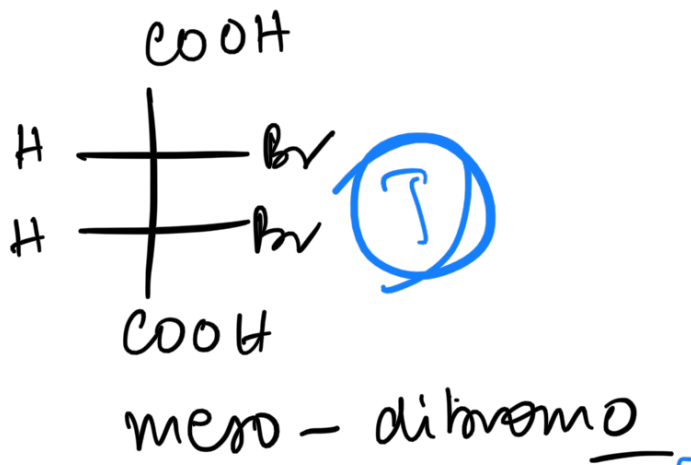
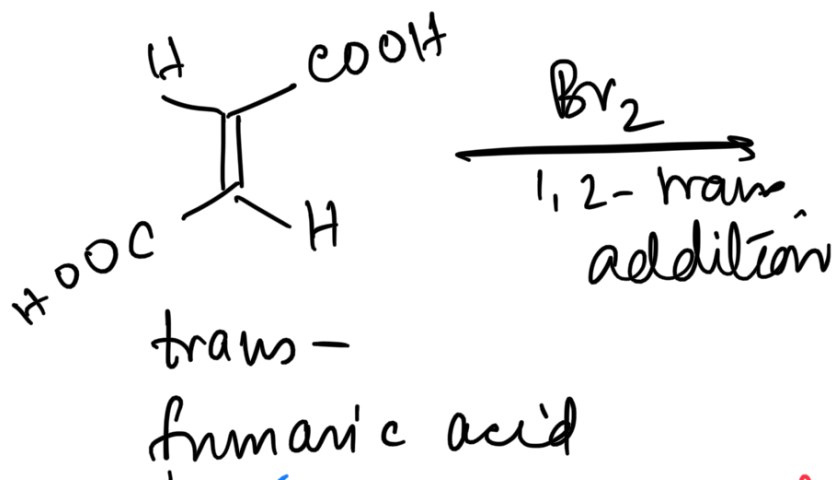
Thiourea

Trapping the intermediate



⑤ Stereochemistry





meso
 (I)